

# YANXIANG ZHU

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## EDUCATION

### Columbia University

New York, NY

*M.S. in Electrical Engineering*

Sep 2026 – Dec 2027

- Selected Coursework: Digital VLSI Circuits, Formal Verification, Analog Electronics Circuits

### Fudan University

Shanghai, China

*B.S. in Microelectronics*

Sep 2022 – Jun 2026

- Overall Score: **92/100**
- Selected Coursework: System on Chip, Digital Integrated Circuit, VLSI Technology, ASIC Design Methodology
- Scholarships: Fudan University First-Class Scholarship (awarded to the top 3% of students), NITORI International Scholarship, Fung Scholarship
- Semester Exchange Student, **University of Hong Kong**, Hong Kong SAR (Aug 2024 – Dec 2024)
- Summer Exchange Student, **University of Texas at Austin**, Austin, TX, USA (Jul 2023 – Aug 2023)

## SKILLS

**Programming Languages:** (Hardware) Verilog, VHDL, Chisel; (Software) Python, C, C++, MATLAB;

**Frameworks & Design Tools:** FireSim, Petalinux, Vivado, ModelSim, Synopsys Design Compiler, Synopsys IC Compiler

## RESEARCH

### Hardware-Software Co-Acceleration for World Action Models

New York, NY

*Research Assistant, Directed by Prof. Zishen Wan*

Aug 2026 – now

- Built a modular WAM testbed supporting interchangeable model architectures for profiling and acceleration.

### Full-Stack Simulation Framework for Intelligent Control Systems

Shanghai, China

*Research Assistant, Directed by Prof. Jun Han*

Jun 2025 – Apr 2026

- Extended UCB's RoSE framework to a customized SoC platform on a Xilinx Alveo V80 FPGA by redesigning host-FPGA communication and system software, enabling closed-loop pre-silicon co-simulation with AirSim and system-level validation of custom accelerator IP for UAV workloads.

### Stochastic Computing Neural Network Accelerator

Shanghai, China

*Leading Researcher, Directed by Prof. Jiang Xiao*

Sep 2023 – Apr 2026

- Developed stochastic-sequence neural networks and bit-stream operators to reduce the arithmetic cost of conventional inference, replacing multiply-accumulate operations with lightweight bit-wise computation.
- Optimized stochastic encoding, nonlinear operators, and memory organization for FPGA deployment, achieving accuracy within 1% of conventional methods without floating-point arithmetic.

## INTERNSHIP

### Mammoth Supercomputing Semiconductor Co., Ltd.

Shanghai, China

*Digital IC Design Intern — AI Accelerator Team*

Apr 2026 – Jul 2026

- Designed the top-level architecture and RTL compute data path for a CNN super-resolution inference ASIC, integrating DDR4, on-chip SRAM, convolution acceleration, and host interfaces into a complete SoC.
- Designed a 128KB dual-port SRAM with ping-pong buffering to hide memory latency, and implemented hardware-controlled sliding-window dataflow for systolic-array convolution.
- Validated super-resolution workloads on FPGA before tapping-out using the TWM compiler toolchain, resolving instruction, memory, and dataflow issues through cycle-level waveform debugging.

## OPEN-SOURCE

### Advanced RISC-V CPU Design on FPGA

Feb 2025 – Aug 2025

- Built a 5-stage pipelined RISC-V CPU with RV32I ISA support with exception handling, branch prediction (BHT+BTB), instruction forwarding, out-of-order issue, and novel memory architecture, validated by Dhrystone tests and real-time UART control using FPGA.